



MICHAEL MUNAVU

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SUMMARY

I am a passionate Elixir and Phoenix LiveView developer on a mission to transform business ideas into powerful web applications and AI Systems. With a proven track record of 16 hackathon victories over 6 countries and an insatiable drive for innovative solutions, I embark on a journey to weave technical excellence.

WORK EXPERIENCE

DockYard, Inc.

Jun 2026 - Present

Elixir / Phoenix LiveView Software Engineer

- Work alongside some of the best Elixir and AI engineers in the world to design and build amazing, production-grade products on the Elixir/Phoenix LiveView stack
- Contribute to full-stack feature delivery, code reviews, and engineering best practices within a globally distributed team

Prosp3r Ltd

Feb 2026 - May 2026

Senior Elixir Engineer · Contract

United Kingdom · Remote

- Design, build, and scale the platform powering Prosp3r, the first interactive candidate mentorship and job board search engine built exclusively for the Elixir, Erlang, and Clojure community
- Own key technical decisions from architecture through deployment across three core services: CV search for hiring clients, a candidate fulfillment mentorship program, and client job postings & consultation
- Collaborate closely with the team to deliver robust, scalable systems that help engineers in the functional programming ecosystem find and grow their careers

GS1 Kenya

May 2025 - Present

Assistant Manager, AI Systems

- Lead Vumbuzi AI, GS1 Kenya's software and AI arm, building internal tools and AI-integrated platforms for GS1 members across Kenya using the Elixir/Phoenix stack
- Lead a cross-functional engineering team delivering production AI platforms for GS1 members, from discovery through deployment and optimization
- Architected Callwisely, a multi-channel AI customer care platform unifying phone, WhatsApp, and website chat into a single dashboard, now used by external clients
- Directed development of an AI-powered aptitude testing platform that auto-generates role-specific assessments and interview-ready reports from a single candidate link
- Oversaw delivery of a hospital management system covering patient records, appointments, billing, and clinical workflows for GHCE, reducing manual paperwork through workflow automation
- Manage the technical roadmap for AI adoption across GS1 Kenya's member network, balancing hands-on architecture decisions with team leadership and project delivery
- Built fault-tolerant systems using Elixir/Phoenix, OTP supervision trees, and real-time LiveView interfaces to ensure reliability across all member-facing platforms

Virgil

Jan 2024 - May 2026

Senior Elixir Engineer

- Deliver enterprise software consultancy building custom solutions across diverse global markets using Elixir/Phoenix LiveView architecture with comprehensive system design and implementation
- Architect scalable in-house startup platforms leveraging Oban for reliable background job processing, queue management with RabbitMQ, and distributed task scheduling across multiple business verticals
- Build fault-tolerant microservices ecosystem using Elixir with OTP supervision trees, Ecto, Phoenix PubSub, and Oban workers for handling complex business workflows and inter-service communication
- Provide full-stack development services with PostgreSQL databases, real-time LiveView and Tailwind interfaces

African Management Institute (AMI)

Oct 2024 - July 2025

Senior Elixir Engineer

- Architected scalable learning management system using Elixir/Phoenix LiveView with real-time course delivery and interactive assessment modules
- Implemented role-based access control with custom Phoenix plugs and Ecto schemas for multi-tenant user management across instructors, students, and administrators
- Built real-time collaborative features leveraging Phoenix Channels and GenServer processes for live course interactions and progress tracking

- Optimized system performance through efficient Ecto query optimization, database indexing, and OTP supervision trees for fault-tolerant deployment

Podii HQ

Sep 2023 - Oct 2024

Senior Elixir Engineer

- Developed sector-specific management platforms for agriculture and education using Elixir/Phoenix LiveView architecture with custom business logic and domain-driven design patterns
- Built scalable Elixir multi-tenant systems leveraging OTP processes, Oban, and Ecto multi-tenancy for handling diverse operational workflows across different industry verticals
- Implemented real-time data synchronization using Phoenix PubSub and LiveView updates for enhanced user experience and operational efficiency tracking
- Led technical mentorship program conducting hands-on Elixir/Phoenix training sessions, code reviews, and backend development best practices for intern onboarding

GS1

May 2022 - Sep 2023

Junior Elixir Engineer

- Engineered scalable ERP backend systems using Elixir/OTP for high-throughput barcode processing and real-time inventory data synchronization
- Built robust barcode parsing pipelines leveraging GenStage for concurrent processing of GS1 standard formats (EAN, UPC, DataMatrix) with fault-tolerant error handling
- Deployed and managed production infrastructure on AWS using EC2, RDS, and S3 for scalable application hosting, database management, and asset storage with automated CI/CD pipelines
- Orchestrated containerized applications using Docker and DigitalOcean Droplets with load balancing and auto-scaling configurations to handle variable traffic loads

Uamuzi

Sep 2021 - March 2022

Senior Elixir Engineer

- Led engineering team in architecting citizen-government communication platform using Elixir/Phoenix LiveView with real-time messaging capabilities and multi-jurisdictional data management
- Designed scalable Elixir microservices architecture leveraging OTP supervision trees and distributed Elixir nodes for handling high-volume civic engagement across multiple government tiers
- Developed robust data processing pipelines using Broadway for handling message queues and event streaming, with Oban for reliable background job processing and scheduled task management
- Architected caching layers using Cachex and ETS tables for high-performance data retrieval, while implementing rate limiting with Hammer to prevent system abuse and ensure fair resource allocation

PROJECTS

CALLWISELY

www.callwisely.ai

CallWisely.ai delivers enterprise-grade AI-powered phone automation through a robust Elixir/Phoenix LiveView, Tailwind platform that ensures 24/7 call handling without dropped connections or lost revenue opportunities. The system utilizes Oban background workers for processing call queues, managing AI agent workflows, and handling real-time telephony integrations with existing business infrastructure. Built on OTP supervision trees for maximum uptime reliability, the platform features intelligent call routing, automated customer service responses, and seamless CRM integrations through Phoenix Channels and WebRTC protocols.

BREE AI

www.breeai.io

Bree AI delivers comprehensive job application optimization through an intelligent Elixir/Phoenix LiveView platform that processes CV enhancement, cover letter generation, and interview preparation workflows. The system utilizes Oban background workers for handling AI model inference, document processing pipelines, and real-time content generation while maintaining responsive user interactions through LiveView components. Built with OTP fault-tolerance for reliable document processing, the platform integrates natural language processing APIs, voice recognition services, and text analysis engines through Phoenix Channels for seamless AI-powered coaching experiences.

NEXUSCALE

www.nexuscale.ai

Nexuscale AI is an enterprise email marketing platform built on Elixir/Phoenix LiveView that automates AI-driven sales processes and campaign optimization. The system leverages Oban workers for email generation pipelines, customer segmentation, and competitor analysis while utilizing OTP supervision trees for reliable delivery infrastructure. The platform integrates machine learning APIs through Phoenix Channels, employs PostgreSQL for contact management, and uses Oban job scheduling for batch processing AI content generation, automated meeting scheduling, and real-time campaign performance tracking.

ADDITIONAL INFORMATION

- **Technical Skills:** Full-Stack Development, Elixir/Phoenix LiveView, React JS, Ash Framework, Oban, Ecto, PostgreSQL, Tailwind CSS, JavaScript, RESTful APIs, Real-time Systems, Database Architecture, System Integration, Scalable Web Solutions, OTP/GenServer, Phoenix Channels, LiveComponents.
- **Languages:** English.